KULLIYYAH OF ENGINEERING

MECHATRONICS SYSTEM INTEGRATION (MCTA 3203)

SEMESTER 2, 2025 / 2026

WEEK 7 : PLC INTERFACING WITH MICROCONTROLLER AND PC OVER ETHERNET / IP (VERSION 2.0)

SECTION 2

GROUP 4

INSTRUCTORS : ZULKIFLI BIN ZAINAL ABIDIN & WAHJU SEDIONO

Date of Submission: 29th April 2026

NO	NAME	MATRIC NUMBER
1	Hannah Irdeena binti Azrulhafiz	2410018
2	Azzam Azhari Noureldein Elamin	1827295
3	Kamarul Zafri bin Kamarul Ariffin	2323189

ABSTRACT

This module focuses on the integration of automation principles with microcontrollers by interfacing a Programmable Logic Controller (PLC). Focusing on developing a Start-Stop control circuit using a ladder diagram designed in OpenPLC Editor and implemented upon an Arduino. The methodology demonstrates the fundamental operations of designing and simulating ladder diagrams for two primary tasks, a timed blinking LED system by using the TON and TOF timer and a Start-Stop control circuit with latching circuit. The physical circuit, consisting of pushbuttons, LEDs, resistors and a breadboard, was built to test and verify the latch operations. In which, the LED responds to the Start and Stop commands, showcasing successful execution of the control logic. This experiment, confirmed with basic PLC programming and an Arduino, provides a scalable and low-cost foundation for implementing industrial logic on consumer-level microcontrollers.

TABLE OF CONTENTS

1.0 INTRODUCTION

The objective of this lab experiment is to bridge industrial automation control and accessible embedded hardware by applying the fundamental control logic of a Start-Stop Control Circuit using the Programmable Logic Controller (PLC) approach assimilated with an Arduino microcontroller. The main purpose of this activity is to attain hands-on experience in the design, simulation and deployment of this critical control program. The main goals are to learn and understand the OpenPLC Editor (an IEC 61131-3 compliant software) to build latching logic using Ladder Diagram (LD), link the variables in the program to the actual input and output pins on the Arduino (like push buttons and an LED), and upload the program to the board to test and confirm that the circuit works as intended.

A PLC is a type of industrial computer built to control systems in a reliable way. It's often programmed using Ladder Diagram (LD), a visual language that works like traditional relay circuits. In this case, the main focus is a Start-Stop control circuit that uses a latching function. This means when the Start button is pressed briefly, the system keeps the LED turned on even after the button is released. It stays on because the program holds its own signal. The LED only turns off when the Stop button is pressed, which is set up to normally allow current through but cuts it when pressed, breaking the circuit. The program is created using the OpenPLC Editor, and an important step is linking the program's variables to the actual Arduino pins so the inputs and outputs work correctly.

According to the Ladder Diagram logic principles and the expected demeanor of the control circuit, a clear hypothesis is definite. The hypothesis states that the developed LD program will be able to control the connected hardware with the specified logic. Temporarily pressing the UP_Start push button will turn the LED ON as well as latching the circuit and will remain until the DOWN_Stop push button is activated, unlatching the circuit and turning the LED OFF.

2.0 MATERIALS AND EQUIPMENTS

a. OpenPLC Editor Software
b. Arduino Board
c. Push Buttons Switches
d. LED
e. Resistors
f. Jumper Wires
g. Breadboard

3.0 EXPERIMENTAL SETUP

1.0 <u>Software Setup</u>

a. Install and open OpenPLC
b. Create a new project and select the Ladder Diagram (LD)

2.0 <u>Create Circuit</u>

a. Define the variables assigned to the LED, Start and Stop push buttons
b. Compile both ladder programs to verify that no errors are present

3.0 <u>Pin Configuration</u>

a. Decide which Arduino pins that will operate as digital inputs and outputs
b. Link each created variable in OpenPLC to its appropriate address, using %IX for inputs and %QX for outputs.

4.0 <u>Uploading the Program</u>

a. Compile the program again to make sure it's error-free
b. Transfer the completed ladder diagram to the Arduino using the "Transfer to PLC" feature

5.0 <u>Building the Circuit</u>

4.0 METHODOLOGY

4.1 Ladder Diagram (OpenPLC Editor)

1. By opening OpenPLC Editor and creating a new project with Ladder Diagram (LD) as the programming language.
2. Set the variables UP_Start, DOWN_Stop, Latch and LED are defined as local boolean types. With each input and output assigned to its respective physical address in Location (e.g. %IX0.0, %IX0.1 and %QX00)
3. The Start-Stop logic is constructed by building the ladder logic using the standard LD components.
    a. Latching Functions:
    - A normally closed (NC) contact for DOWN_ Stop is in series with a normally open (NO) contact for UP_Start.
    - Add a parallel NO contact to the Latch coil to serve as a seal-in-path and end with it.

    b. Output Activation
    - A NO contact controlled by the Latch.
    - An output device for the rung is LED coil mapped to %QX0.0
4. Ladder logic is compiled to check for any errors.
5. Simulation tool used to test the logic, to confirm latch and LED remain active after pressing the UP_Start button and only turning off when the DOWN_Stop button is initiated.

4.2 Uploading Program to Arduino

1. Connect the Arduino Uno to the computer via USB using the correct COM port.
2. To transfer the code after selecting Arduino Uno as the board type use the "Transfer to PLC" function to load the compiled OpenPLC Runtime program onto the Arduino board.

5.0 Data Collection

Test Sequence	Start Button	Stop Button	Expected LED	Observed LED	Latch Status
1	Not pressed	Not pressed	OFF	OFF	Not latched
2	Pressed	Not pressed	ON	ON	Latched
3	Released	Not pressed	ON(Stays)	ON(Stays)	Latched
4	Not pressed	Pressed	OFF	OFF	Unlatched
5	Not pressed	Released	OFF(Stays)	OFF(Stays)	Not latched
6	Pressed	Not pressed	ON	ON	Latched
7	Pressed	Pressed	OFF	OFF	Unlatched (Stop overrides)
8	Released	Released	OFF	OFF	Not latched

6.0 Data Analysis

The Start-Stop circuit data confirms the correct implementation of the self-latching (seal-in) pattern:

1. **Latching Behavior:** Once the Start button is pressed and released (Tests 2,3), the LED remains ON because the parallel MotorCoil contact maintains the current path. This is the core function of the latch in industrial motor control.
2. **Stop Priority:** Test 7 demonstrates a critical safety feature when both Start and Stop are pressed simultaneously, the output turns OFF. This is because the NC Stop contact is in series with the entire circuit, so it breaks the current path regardless of the Start contact state. In industrial applications, this ensures that emergency stop conditions always override start commands.
3. **Response Time:** The measured response time of approximately 25ms corresponds to one or two scan cycles of the OpenPLC Runtime on the Arduino Mega. This is sufficiently fast for manual button-press applications, though industrial PLCs typically achieve sub-millisecond response times.

7.0 Results

- The Start Stop ladder diagram with parallel latch contact and series NC Stop contact compiled and simulated correctly.
- All eight test sequences produced the expected results on both simulation and hardware.

- The self latching behavior was confirmed: pressing Start turned the LED ON, and releasing Start kept it ON.
- The Stop button successfully broke the latch and turned the LED OFF.
- Stop priority over Start was verified when both buttons were pressed simultaneously.
- The physical circuit with pushbuttons on D22/D23 and LED on D14 responded identically to the simulation.

8.0 DISCUSSION

The outcome of this experiment shows the proper operation of a fundamental industrial Start-Stop control circuit using the OpenPLC Editor with an Arduino Mega. The observed changes in the system matched what was expected from the Ladder Diagram (LD), showing that the OpenPLC runtime correctly turned the latching logic into real input and output actions on the Arduino Mega's pins. When the Start button was pressed briefly, the LED turned on and stayed on even after it was released, proving that the latching function worked properly. In the same way, pressing the Stop button which was set up as a normally closed contact, immediately broke the circuit and turned the LED off just as intended.

The following results suggest that the Arduino Mega functions as an educational and cheap alternative PLC platform. The Arduino Mega has more input and output pins, which makes it especially useful for learning and building projects that involve multiple inputs and outputs. This experiment shows that basic PLC ideas like latching, normally open (NO) and normally closed (NC) behavior, and state-based logic can be effectively recreated on a microcontroller using the OpenPLC system.

There were no significant inconsistencies that arose between expected and observed behavior. Although minor issues like button bounce can occur in theory, their effect was minimal in this case due to the relatively slow scan cycle and the simplicity of the digital logic used. Whereas the system operated uniformly with the standard PLC ladder logic principles.

9.0 CONCLUSION

The module successfully demonstrated the integration of the Programmable Logic Controller (PLC) environment with a microcontroller through OpenPLC Editor. By illustrating the usage of standard function (timer block) to regulate the interval of a blinking LED in a ladder diagram, specifically TON and TOF, the system proves to manage precise timing operations, which is essential for controlling the automated industry.

Developing the Start-Stop control circuit involves using the latching logic to maintain an output state through a “Latch” bit instruction. The result of the task was a successful mapping of software variables to physical hardware using standardized addressing to pushbuttons. This is further verified with the physical circuit confirming the ladder logic was executed correctly on the Arduino. In conclusion, this confirms that basic PLC programming provides a scalable and low-cost foundation for real-world industrial logic on accessible microcontroller platforms.

10.0 RECOMMENDATIONS

To make the system more practical in the future, it is suggested to replace the simple LED with a relay module or a motor driver. This change would allow the Arduino Mega to operate real mechanical parts, providing a much more realistic example of how controllers move physical equipment. Additionally, adding safety features like an emergency stop or an overload switch would be a great way to improve the project. This would help demonstrate how safety rules in the software can automatically shut down a machine, even if a "start" command is being given, which is a vital part of industrial safety.

The experiment also highlighted several key lessons regarding how hardware and software work together. One of the most important steps is ensuring that the input and output pins are mapped correctly in the code so that the signals go to the right places. It is also essential to understand how physical buttons, whether they are normally open or normally closed, translate into the logic used by the controller. Mastering these basics is a fundamental skill for anyone looking to design control systems that are both reliable and safe for everyday use.

11.0 REFERENCES

1. Control.com. PLC Ladder Logic on an Arduino: Introduction to OpenPLC. Available at: https://control.com/technical-articles/plc-ladder-logic-on-an-arduino-introduction-to-openplc/
2. Autonomy Logic. OpenPLC Editor. Available at: https://autonomylogic.com/

12.0 APPENDICES

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to Dr. Zulkifli Bin Zainal Abidin and Dr. Wahju Sediono for their continuous support and for providing the essential resources and facilities required to carry out this project. We also extend our heartfelt appreciation to everyone who contributed to the success of this lab report through their valuable insights, assistance, and feedback.

STUDENTS DECLARATION

We hereby affirm that the work presented in this report is entirely our own. All sources used for background information and theory have been properly cited, and have strictly adhered to the principles of academic integrity throughout this report.